HyperMLP: Superpixel Prior and Feature Aggregated Perceptron Networks for Hyperspectral and LiDAR Hybrid Classification

Jiaojiao Li, *Senior Member, IEEE*, Yuzhe Liu, Rui Song, *Member, IEEE*, Wei Liu, Yunsong Li, *Member, IEEE*, and Qian Du, *Fellow, IEEE*

***Abstract*— Hyperspectral images (HSIs) have excellent spectral combining capabilities and light detection and ranging (LiDAR) images have fine stereoscopic elevation information. Therefore, the multimodal fusion classification of hyperspectral and LiDAR images inevitably improves the interpretation ability of remote sensing images. In recent years, the MLP-Mixer, an image processing network based on multilayer perceptron (MLP), has flourished in the field of image processing. In this work, we propose an innovative HyperMLP network based on the deep learning framework MLP-Mixer architecture to address the lack of spatial feature construction capability and the locality of multimodal feature fusion in naive networks. Specifically, first, the adoption of unsupervised superpixel embedding provides additional shallow morphological spatial feature information for the network, reduces the pressure of the feature extraction network, and enhances feature discrimination capabilities. Second, the feature scrambling strategy improves the diversity of features and strengthens the generalization of the network by enhancing interactions between different spatial features. Finally, by implementing the bilateral modulation strategy, feature fusion is applied at every stage of the deep network, reducing semantic drift between features. On three fiducial remote sensing datasets, classification tests are performed on the proposed HyperMLP network to verify its performance, and the results are definitely impressive.**

Manuscript received 10 December 2023; revised 27 December 2023; accepted 12 January 2024. Date of publication 17 January 2024; date of current version 31 January 2024. This work was supported in part by the National Nature Science Foundation of China under Grant 62371359, in part by the Fundamental Research Funds for the Central Universities under Grant JBF220101, in part by the Key Research Program of the Chinese Academy of Sciences under Grant KGFZD-145-2023-15, in part by the State Key Laboratory of Geo-Information Engineering under Grant SKLGIE2020-M-3-1, in part by the Wuhu and Xidian University Special Fund for Industry–University–Research Cooperation under Grant XWYCXY-012021002, in part by the 111 Project under Grant B08038, and in part by the Youth Innovation Team of Shaanxi Universities. *(Corresponding authors: Rui Song; Wei Liu.)*

Jiaojiao Li is with the State Key Laboratory of Integrated Service Networks, Xidian University, Xi'an 710071, China, and also with the CAS Key Laboratory of Spectral Imaging Technology, Xi'an 710119, China (e-mail: jjli@xidian.edu.cn).

Yuzhe Liu, Rui Song, and Yunsong Li are with the State Key Laboratory of Integrated Service Networks, Xidian University, Xi'an 710071, China (e-mail: 21011210556@stu.xidian.edu.cn; rsong@xidian.edu.cn; ysli@mail.xidian.edu.cn).

Wei Liu is with the Key Laboratory of Smart Earth, Beijing 100094, China, and also with the State Key Laboratory of Geo-Information Engineering, Wuhan University, Wuhan 430072, China (e-mail: 1271772072@qq.com).

Qian Du is with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS 39762 USA (e-mail: du@ece.msstate.edu).

Digital Object Identifier 10.1109/TGRS.2024.3355037

***Index Terms*— Hybrid modality classification, hyperspectral image (HSI), light detection and ranging (LiDAR), MLP-Mixer.**

I. Introduction

HYPERSPETRAL images (HSIs) contain extensive spatial information and hundreds of continuous spectral bands, which enable adequately quantitative analysis of the material and spatial properties of targets. Over the last few years, HSIs have been applied to a wide range of applications, including remote sensing, industrial production, mineral exploration, and medicine, to name a few [1], [2], [3], [4]. The classification of hyperspectral imagery is an essential technology for interpreting hyperspectral data, which further expands the application prospect. However, the classification accuracy might be limited by the fact that data from a single sensor cannot provide comprehensive and rich features of classes to accurately discriminate the pixels in imagery [5]. Currently, the most effective approach is to combine and complement multiple image sensors, including RGB images, synthetic aperture radar (SAR), multispectral images, panchromatic images, light detection and ranging (LiDAR), and others [6], [7], [8]. In particular, LiDAR-derived digital elevation model (DEM) is a digital simulation of ground topography enabled by limited terrain elevation data. It is a kind of physical ground model that represents the ground elevation in the form of a set of ordered numerical arrays, which can effectively make up for the shortcomings of hyperspectral sensors [9]. Therefore, the fusion between LiDAR and HSIs for further classification has received extensive research and attention in the current academic field [10]. A typical application scenario, such as a concrete pavement and a concrete roof, has almost the same reflectance spectrum, but after combining the elevation information, they are relatively easy to distinguish.

Over the past few years, convolutional neural networks (CNNs) have become increasingly powerful at extracting features from image data and CNN-based hyperspectral fusion classification networks have also been rapidly developed [11]. For instance, in the fusion classification task, Xu et al. [12] proposed a dual-branched feature extraction network that extracts hyperspectral and LiDAR features simultaneously, maintaining the advantages of both branches at once. On the basis of the dual-branch network, Hang et al. [13] proposed

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:25:56 UTC from IEEE Xplore. Restrictions apply.

the couple CNN for HSI classification, which incorporates the characteristics of Siamese networks into the feature extraction process. Nevertheless, the restricted receptive fields of the convolution kernel prevent the traditional deep networks from fully capturing multilevel and multiscale features. Also, the most rational approach to solving the above problems is to employ feature extractors of different sizes for fusion and attention methods for separating the features. Typically, the FusAtNet [14] achieved superior performance in fusion classification by adding multiscale features as well as the attention module to the network and by combining both strategies to achieve superior performance in fusion classification. While the above network achieved excellent performance due to its exquisitely designed deep network architecture, it still cannot overcome the inherent deficiencies of convolutional networks. These limitations are typically such as the fact that convolutional networks only process data based on local feature models and are unable to access the entire map or global interactions between features, which results in a lack of generalization and globalization of the feature extraction network, particularly in hyperspectral classification task, resulting in interclass confusions [15].

To solve the above inherent dilemma of CNN networks, a novel transformer architecture based on attention has been devised as a result of the enhancement of the attention mechanism, making a qualitative advance in natural language processing [16]. Aside from that, the vision transformer (ViT) performed impressively in image processing [17]. A revolutionary paradigm for HSI classification was proposed by SpectralFormer [18] with the ViT architecture for HSI processing, allowing the feature extractor network to obtain a global view and improving the performance of the hyperspectral classification network. In addition, an innovatively developed feature extraction algorithm called DHViT [19] pairs HSIs with LiDAR images, utilizing convolutional and transformer networks with both local and global receptive fields. While the aforementioned networks have achieved remarkably excellent classification results, they still confront several challenges. More specifically, hyperspectral data are characterized by high dimensionality, which leads to a computationally intensive combination of transformer and CNN networks with heavy operating parameters, which is largely limited by the hardware. In addition, subsequent studies have found that the image data in the transformer structure have a certain amount of information redundancy, which indicates the need for a network that effectively reduces the number of parameters in order to reduce the hardware requirements [20], [21].

Lately, recent studies have proposed a novel multilayer perceptron (MLP) feature perceptron network that achieves impressive fusion classification performance through feature embedding and fully connected layers [22]. Compared to conventional transformers, the MLP network is designed similar to a transformer, but with a lighter and more compact structure than the transformer. Specifically, the MLP architecture consists of consecutive token mixer modules and channel mixer, which are connected by feature transposition, and combining multiple modules to realize feature interaction in different dimensions. For HSI classification tasks, the channel mixer and marker mixer in the MLP architecture are well suited to simultaneously integrate HSIs and process spatial and spectral information [23], [24]. In the meantime, several different network architectures have been developed based on the MLP architecture. Typically, to enhance robustness, ResMLP [25] employed affine transformations to embed features and added linear feature mapping to avoid the performance impact of feeding images directly to the MLP feature extractor. The spectral–spatial MLP (SSMLP) [26] proposes a combined spatial–spectral MLP architecture for HSI classification, which improves the ability of MLP to model spatial–spectral features of HSIs.

Benefiting from the above superior performance of the MLP-Mixer-based network, we propose a novel HyperMLP applied to the fusion of hyperspectral and LiDAR classification tasks via adopting the MLP architecture as the primary architecture in this article. Specifically, the network is divided into three feature extraction branches to extract spatial, spectral, and elevation features of hyperspectral and LiDAR images. A novel feature shuffle module (FSM) is proposed in the LiDAR feature extraction branch to perform feature reorganization, which combines spatial location with cross-reorganization to interact with each embedding set. At the same time, a feature recovery process is implemented to ensure that the spectral features in the mixer part of the subsequent channels are not corrupted. This module enhances the joint modeling capability of the MLP architecture for spatially global features in elevation imagery, increases the information interaction of features at different locations, and further improves the classification accuracy by increasing the feature diversity. In addition, to further enhance the shallow spatial feature extraction capability of the MLP network, we utilize the outputs of the unsupervised superpixel segmentation as additional embedding data to enhance the morphological feature representation. By adopting this strategy and combining it with the MLP network to extract spatial features, shallow spatial clustering embedding information can be obtained from the input data, which reduces the pressure of spatial feature extraction and improves the convergence speed and classification effect of the network. Aside from that, we propose a bilateral modulation strategy (BMS) to monitor the multimodal data throughout the network training process in order to avoid fragmentation of the feature extraction. More specifically, for each mixing block of the two feature extraction branches, the features of both branches are aligned using affinity matrices to reduce the intraclass variance in feature extraction for different modal data. Furthermore, the affinity matrix is updated as the training iterations increase. Finally, the different modal features are bilaterally modulated, thus constraining the feature extraction results in order to more efficiently transfer the preextraction features to the subsequent stages. Overall, the main contributions of this article can be summarized as follows.

1) In this article, we present a novel HyperMLP architecture that employs for the first time a hierarchical MLP architecture for fusion classification of hyperspectral and LiDAR data. The MLP network has a rather concise

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:25:56 UTC from IEEE Xplore. Restrictions apply.

network structure, thus avoiding redundant matrices and attention calculations while achieving competitive classification results on three benchmark datasets.

2) HyperMLP performs unsupervised preprocessing of hyperpixels and groups spatially similar pixels by entropy rate strategy, which acquires a priori shallow morphological information and enhances the perceptual ability of MLP network to extract spatial features from hyperspectral and LiDAR data.
3) We introduce a FSM in the mixer for transforming and combining different features. This module overcomes the lack of spatial awareness of MLP networks, enriches the feature diversity, improves the network's interaction between different feature embeddings, and further increases the robustness of MLP networks.
4) A BMS is proposed in this article for multimodal fusion of hyperspectral and LiDAR images. A variable affinity matrix is updated iteratively between different stages, resulting in two branch networks learning features in a complementary manner, avoiding fragmentation and fragmentation of feature learning, and increasing the representation capabilities of fusion features.

II. Related Works

A. MLP Network

In recent years, CNN-based networks have made a big splash in computer vision, and transformer architectures have become increasingly powerful, but the potential drawbacks of both have come to light. Recent studies have shown that convolutional and attention mechanisms are not indispensable in image perception, and MLP-based architectures have achieved competitive results on large datasets [22]. On this basis, ResMLP created residual-like structures based on the original network [25] and then made the patches interact with each other through linear layers. Guo et al. [27] proposed HireMLP, which made use of cross-region feature rearrangement of spatial and channel dimensions to achieve information communication at different locations and obtain excellent accuracy for object detection tasks. In the field of hyperspectral classification, many works also adopted MLP architectures. Shao et al. [28] made improvements based on CycleMLP [23], expanded the feature perception range of MLP, improved the MLP network from the spatial and spectral interaction, and improved the classification performance. Tang et al. [29] applied the idea of residual networks to HSI classification, combined with MLP network architecture, and obtained excellent classification results. In addition, multiscale U-shape MLP (MUMLP) adopts the classical U-shape encoder–decoder network structure for feature extraction [30] with MLP-Mixer as the encoder backbone, and excellent classification performance is obtained on multiple datasets. However, in the MLP network proposed for hyperspectral classification, there is no research on using LiDAR elevation information for fusion classification. This article is also the first time that the MLP-Mixer architecture is applied to the multimode fusion classification of hyperspectral and LiDAR.

B. Superpixel Segmentation

The superpixel technique composes a series of small regions of neighboring pixels with similar characteristics (e.g., brightness and texture). These small regions contain valid information from the segmentation process and retain most of the boundary texture information. In HSI processing tasks, superpixel segmentation is extensively utilized as a preprocessing step in algorithms [31]. The most classical method for superpixel segmentation is simple linear iterative clustering (SLIC) [32]. First, the initial seed is randomly selected, and the gradient value of all pixels in the neighborhood is calculated, and then, the superpixel result is obtained by iterative clustering. The distance is calculated as follows:

$$d_c = \sqrt{(l_j - l_i)^2 + (a_j - a_i)^2 + (b_j - b_i)^2} \tag{1}$$

$$d_s = \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2} \tag{2}$$

$$D' = \sqrt{\left(\frac{d_c}{N_c}\right)^2 + \left(\frac{d_s}{N_s}\right)^2} \tag{3}$$

where d_c represents the color distance, d_s represents the spatial distance, N_s is the maximum spatial distance within the class, and N_c is the maximum color distance, all of which are characteristic parameters for images and clustering. In addition, the graph-based entropy rate superpixel (ERS) segmentation method is also widely employed [33], which introduces a linear independent combination structure in the generalized vector space to maximize the objective function under the constraint of the graph topology. For the sake of computational efficiency, this article also adopts this superpixel method. In addition, density-based mean-shift method [34], region-based turbopixels method [35], superpixels extracted via energy-driven sampling (SEEDS) method based on gradient ascent [36], and superpixel segmentation methods based on linear spectral clustering (LSC) algorithm [37] are also widely utilized.

C. Hyperspectral and LiDAR Fusion Classification

Recently, due to the spectral instability of HSIs, more and more studies tend to use other sensor data as an additional prior to improve the accuracy of the HSI processing [9], [38], [39]. Typical ones are SAR, LiDAR, panchromatic images, and so on. In the HSI classification, the same object has different spectra. The LiDAR elevation model with fine height information can effectively make up for this defect [10]. To improve the pertinence and effectiveness of feature extraction, two-branch CNN adopts a unique two-channel network to perform feature perception on LiDAR features and spatial–spectral features [12]. In addition, multiscale and channel attention methods are also effective solutions to improve task accuracy. FusAtNet [14] proposes an effective multiscale cross-layer feature fusion scheme to enhance the interaction of features between different layers and different scales. However, the blind superposition of multilayer features will also affect the performance of the network. Accordingly, Li et al. [10] proposed a cross-layer feature enhancement scheme, which

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:25:56 UTC from IEEE Xplore. Restrictions apply.

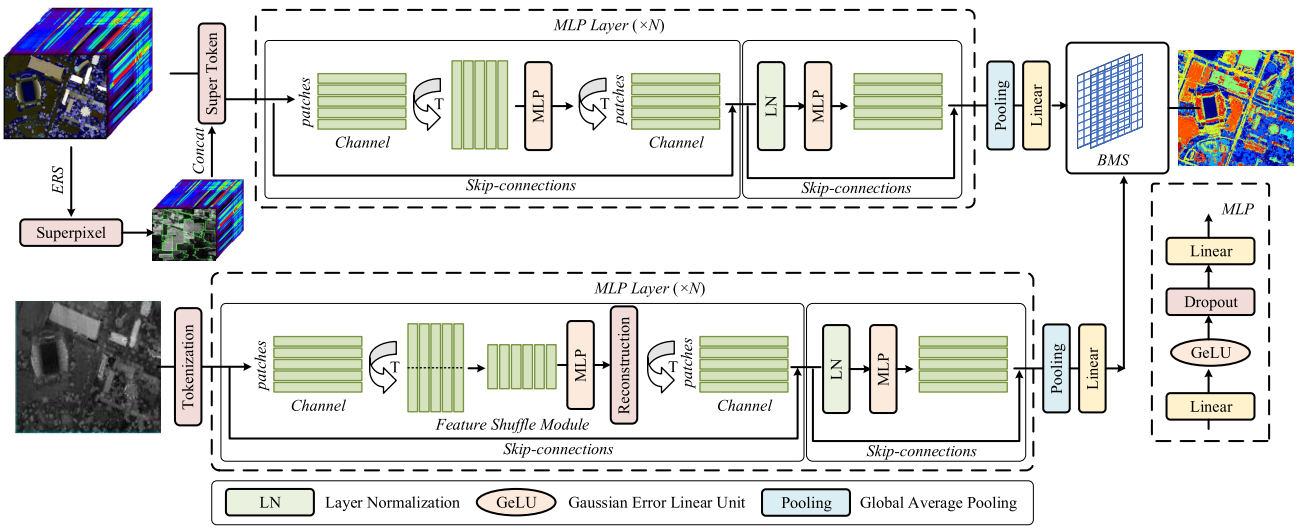


Fig. 1. Overall of the proposed framework. It includes the superpixel segmentation prior information "Super Token," the LiDAR feature shuffle module "FSM," and the feature fusion "BMS." "T" in the figure represents the feature dimension transpose. "ERS" stands for entropy rate superpixel method.

effectively solved the problem of feature redundancy and improved the classification accuracy of HSIs. With the emergence of the Transformer architecture, which is also used in this task, SpectralFormer [18] is the first to use the ViT structure for HSI classification and also designs a unique grouping input to increase the diversity of transformer tokens. DHViT [19] applies the dual-branch idea to the ViT structure and designs a method combining convolution and transformer to process HSI and LiDAR fusion images. For hyperspectral classification under the MLP architecture, although there have been previous works that deal with a single HSI, this article is the first to use the MLP-Mixer architecture for multimodal fusion classification of HSI and LiDAR images.

III. Proposed Method

This section provides a detailed description of the proposed HyperMLP architecture. First, the overall network is introduced, followed by an elaborate description of each component within the network.

A. Superpixel-Based MLP Mixer Architecture

Fig. 1 shows the overall network flow, which includes the MLP Mixer processing part for HSIs and the extra MLP Mixer for extracting LiDAR elevation information, and the BMS for multifeature fusion.

Different from the direct input processing method, we preprocess the input hyperspectral data with superpixels and aggregate the spatially similar pixels before feature extraction, which mitigates the disadvantage of the MLP Mixer network in spatial feature perception.

For the HSI cube of size $\mathbf{X} \in \mathbb{R}^{H\times W\times B}$, where H, W, and C represent the length and width of the patch and the number of bands, respectively. In order to obtain the preliminary spatial feature projection of the same pixel region, it is first necessary to reduce the dimensionality of the HSI data with high dimensions, and we used the principal component analysis (PCA) dimensionality reduction method for processing. PCA method is a common data processing method, which can extract the main feature components along the channel dimension of HSIs. The original data $\mathbf{X}$ are first deformed and transposed into a 2-D matrix $\mathbf{X}_t \in \mathbb{R}^{HW\times B}$. Also, the input data can be represented as

$$\mathbf{X}_t = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_B]^T \tag{4}$$

where $\mathbf{x}_i$ $(1 \le i \le B)$ represents the information of each band. At the same time, a low-dimensional feature space $\mathbf{L} \in \mathbb{R}^{HW\times D} = [\mathbf{l}_1, \mathbf{l}_2, \ldots, \mathbf{l}_D]^T$ is generated according to the number of bands D in the target dimension reduction. For each subband, the linear transformation matrix $\mathbf{M}$ is defined such that the following condition is satisfied:

$$\mathbf{L} = \mathbf{M}^T \cdot \mathbf{X}_t. \tag{5}$$

Mathematically, the covariance matrix of the feature $\mathbf{X}_t$ is first solved, and then, the eigenvector is found by the Lagrange multiplier method. The first D is taken as the desired $\mathbf{M}$ according to the target band

$$\mathbf{C} = \frac{1}{B}(\mathbf{X}_t - \theta_i)(\mathbf{X}_t - \theta_i)^T \tag{6}$$

$$\mathbf{M} = \max_{\text{top}D}\left(\frac{1}{B}(\mathbf{X}_t - \theta_i)(\mathbf{X}_t - \theta_i)^T\right) \tag{7}$$

where $\mathbf{C}$ is the covariance matrix of the input features and $\mathbf{M}$ is the transformation matrix. Then, the spatial structure of the $\mathbf{L}$ after PCA is restored and the ERS is performed.

The ERS method applies the entropy rate as the objective function, while the graph random walk method is used to perform spatial pixel aggregation. In general, the entropy rate concept is denoted by

$$\mathcal{H}(\mathbf{S}) = \lim_{t\to+\infty} H(S_t|S_{t-1}, S_{t-2}, \ldots, S_1) \tag{8}$$

where $\mathbf{S} = \{S_t|t \in T\}$ denotes the uncertain stochastic process and T is the index set. In ERS, the HSI data are represented by a directed graph, and the random walk algorithm is used

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:25:56 UTC from IEEE Xplore. Restrictions apply.

to transform the original data. We take the feature of the first channel after dimension reduction for superpixel because it contains the most abundant information, denoted as $\mathbf{X}_f$, and then, the final superpixel segmentation result is as follows:

$$\mathbf{X}_{\text{ers}} = \bigcup_{k}^{n} \mathcal{X}_k, \quad \text{s.t. } \mathcal{X}_k \cap \mathcal{X}_g = \emptyset \tag{9}$$

where n represents the superpixels numbers and $\mathcal{X}_k$ is the kth superpixel.

After that, the results of superpixels are concatenated with the original hyperspectral data as an additional channel, and the new data input is

$$\mathbf{X}' \in \mathbb{R}^{H\times W\times(B+D)} = \text{concat}(\mathbf{X}, \mathbf{X}_{\text{ers}}). \tag{10}$$

After the patch division of the synthesized new data, token embedding is performed as the input of the MLP-Mixer architecture. The result of the embedding is a 2-D block of scale $S \times C$, where S divides the patch (P, P) according to the original pixel (H, W) of the image, and then,

$$S = HW/P^2. \tag{11}$$

In particular, for HSIs that originally take patches as input, we set $P = 1$ to accurately model the interspectral information. Also, the piecewise embedded image is denoted as $\mathbf{X}^h \in \mathbb{R}^{S\times C}$. Later, MLP-Mixer block includes spatial mixer and spectral mixer, and each part performs similar feature mapping

$$\text{spatial mixer: } \mathbb{R}^S \to \mathbb{R}^S \tag{12}$$

$$\text{spectral mixer: } \mathbb{R}^C \to \mathbb{R}^C. \tag{13}$$

For each mixer, which consists of a fully connected network layer composed of Multi-Layer Perceptrons (MLPS) and activation functions, the specific operations can be written as follows:

$$\mathbf{Z}_{*,i} = \mathbf{X}^h_{*,i} + \sigma_2\big(\sigma_1 \text{LN}\big(\mathbf{X}^h_{*,i}\big)\big), \quad \text{for } i = 1, \ldots, C \tag{14}$$

$$\mathbf{Y}_{j,*} = \mathbf{Z}^h_{j,*} + \sigma_4\big(\sigma_3 \text{LN}\big(\mathbf{Z}^h_{j,*}\big)\big), \quad \text{for } j = 1, \ldots, S \tag{15}$$

where LN stands for layer normalization and σ stands for fully connected operation. It is worth noting that the classical activation function rectified linear unit (ReLU) is replaced by element-wise nonlinearity [Gaussian Error Linear Unit (GeLU)] in mixer. At the same time, in order to solve the problem that the fully connected layer cannot model across space, the feature transpose is performed between the spatial mixer and the spectral mixer to interact with different dimensions of information.

B. Features Shuffle Module

For the elevation of DEM information in the LiDAR image, since it is a single-channel image, we design a separate LiDAR Multi-Layer Perceptrons (LiMLP) layer for feature extraction. Unlike CNN, MLP lacks feature extraction between different spaces, and the receptive field is limited to extract only local features. To solve this problem, we modify the feature flow in mixer and add a FSM before feature extraction, which shuffles the input LiDAR embedding along the spatial feature direction and concatenates the shuffled feature vectors along the channel dimension. Assuming that the input feature is $\mathbf{X}_l \in \mathbb{R}^{S\times C}$, we randomly select half of the spatial inputs for the concatenation operation, which can be described as follows:

$$\mathbf{X}'_l = [\mathcal{X}_i] \quad (i \in (1, S), \text{num}\{i\} = S/2) \tag{16}$$

where $\mathcal{X}_i$ represents the randomly selected features, and the number is half of all features. The randomly selected features are then concatenated with the remaining features along the channel dimension, and the concatenated features can be expressed as

$$\mathbf{X}_{\text{Li}} \in \mathbb{R}^{\frac{S}{2}\times 2C} = \mathbf{X}'_l \| \big(\mathbf{X}_l \backslash \mathbf{X}'_l\big). \tag{17}$$

The symbol $\|$ denotes the feature concatenation operation and the symbol $\backslash$ denotes the difference set of the two sets.

Compared with the original feature, the corrected feature is the concatenation of the current feature and another random feature in the channel dimension. After MLP feature extraction, the interaction between the current pixel embedding and another random pixel embedding can be realized. After each MLP-Mixer block, the random feature points are reconstructed and the same random operation is performed in the next round. The spatial MLP at this point can be formulated as follows:

$$\mathbf{X}'_{\text{Li}} = \mathbf{X}_{\text{Li}} + \sigma(\sigma \text{LN}(\mathbf{X}_{\text{Li}})), \quad \mathbf{X}_{\text{Li}} \in \mathbb{R}^{\frac{2}{S}\times 2C}. \tag{18}$$

The proposed FSM module effectively overcomes the lack of interaction between features in the traditional MLP network, adds communication between pixels in the input patch, and improves the robustness of the network. At the same time, it increases the uncertainty of feature extraction and improves the generalization of feature extractor.

C. Bilateral Modulation Strategy

In multimodal data fusion, the proposed HyperMLP designs additional branches for feature extraction separately. This scheme can extract specific features for each type of sensor to improve the identifiability and accuracy of the features. However, for the data of different sensors, the sensor perception is different, and the characteristic deviation and drift of different data sources will exist in the gradually deepening network. Traditional fusion methods are mostly direct feature splicing and assembly, which cannot effectively avoid this problem. Therefore, the bilateral modulation module proposed in this article aims to modulate and correct different sensors through the features learned at each block. Specifically, we use the features of the HSI and the elevation features of the LiDAR image to learn a similarity matrix and then use this matrix as a weight to increase the high similarity part in the feature extraction, so as to represent the common information in the multimodal data features. However, for the information with different representation, that is, the part with low similarity, we give a lower weight to constrain, so as to avoid the interference of this confusing information to the final classifier.

The overall flowchart of the proposed BMS is shown in Fig. 2.

As shown in Fig. 2, after the HSI feature $\mathbf{F}^h_t$ and LiDAR feature $\mathbf{F}^l_t$ of different stages are extracted by

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:25:56 UTC from IEEE Xplore. Restrictions apply.

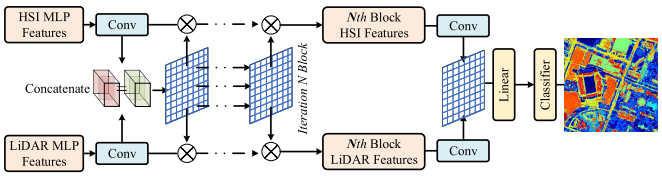


Fig. 2. Flowchart of the BMS. The update process of the similarity matrix was included, and the matrix was updated after each block, and the training of the next block was guided.

the dual-branch network, they are concatenated along the channel dimension. Then, the standard convolution is used for similarity extraction, and the sigmoid function output probability matrix is used as the similarity matrix of the current stage of the two

$$\mathbf{Aff}_t = \Gamma\left(\xi\left(\mathbf{F}_t^h \middle\| \mathbf{F}_t^l\right)\right) \tag{19}$$

where Γ stands for the sigmoid function that outputs the probability map and ξ stands for the standard convolution output.

The obtained feature affinity matrix map is multiplied with each sensor feature as the correction parameter in the next stage. Also, the specific process can be described by the following formula:

$$\mathbf{F}_{t+1}^h = \zeta_1\left(\mathbf{Aff}_t \times \mathbf{F}_t^h\right) \tag{20}$$

$$\mathbf{F}_{t+1}^l = \zeta_2\left(\mathbf{Aff}_t \times \mathbf{F}_t^l\right) \tag{21}$$

where ζ represents one MLP-Mixer feature extraction. It is worth noting that after the last block, the calculated similarity matrix no longer modulates the two branches separately but jointly adjusts the final output of the two branches and then outputs the final classification result through the classification head. ϕ represents the final classifier, and the process can be described as

$$\text{output} = \phi\left(\mathbf{Aff}_n \times \left\{\mathbf{F}_n^h \middle\| \mathbf{F}_n^l\right\}\right). \tag{22}$$

IV. Experimental Results and Analysis

A. Benchmark Datasets

In this article, we employ three generic HSIs with their corresponding LiDAR datasets.

1) *Houston Data:* The Houston data that were used in the 2013 GRSS Data Fusion Competition (DFC) were a competition dataset. It was collected from the University of Houston campus and nearby urban areas. In addition to hyperspectral imagery, a LiDAR digital surface model (DSM) was included. The Houston dataset images had a spatial resolution of 2.5 m and an image size of 349 × 1905 pixels and contained 144 consecutive spectral bands that were distributed between wavelengths of 380 and 1050 nm. The dataset consisted of a total of 15 029 finely labeled data points divided into 15 categories. In addition, the dataset was accompanied by a detailed description that is shown in Table I.
2) *TRENTO Data:* The dataset used was taken in a suburban area of the city of Trento, Italy. The HSI had a spatial resolution of 1 m, was 166 × 600 in size, contained 63 bands, existed in the range of 420–990 nm, and consisted of 30 214 finely labeled sample points divided into six classes. A detailed description of the dataset is provided in Table II.
3) *MUUFL Gulfport Data:* The data were captured at the University of Mississippi Bay Park. The HSI had a spatial resolution of 1 m and contained 64 consecutive spectral bands distributed between wavelengths of 380 and 1050 nm. The data consisted of a total of 53 732 finely labeled sample points that were divided into 11 specific object classes. It is worth noting that 100 samples were randomly selected from each class as the data training set, while the remaining sample points were used as the test set. A detailed description of the dataset categories is provided in Table III.

TABLE I
Categories and Sample Numbers for the Houston Data

Class		Samples Number	
No.	Name	Train	Test
1	Health grass	198	1053
2	Stressed grass	190	1064
3	Synthetic grass	192	505
4	Trees	188	1056
5	Soil	186	1056
6	Water	182	143
7	Residential	196	1072
8	Commercial	191	1053
9	Road	193	1059
10	Highway	191	1036
11	Railway	181	1054
12	Parking lot1	192	1041
13	Parking lot2	184	285
14	Tennis court	181	247
15	Running track	187	473
Total		2832	12197

TABLE II
Categories and Sample Numbers for the TRENTO Data

Class		Samples Number	
No.	Name	Train	Test
1	Trees	129	3905
2	Buildings	125	2778
3	Ground	105	374
4	Woods	154	8969
5	Vineyard	184	10317
6	Roads	122	3525
Total		819	29395

B. Experimental Results

We are in Houston, MUUFL Gulfport along, and TRENTO compared with algorithm based on the different datasets, to verify the effectiveness of the proposed algorithm. Specifically, among the CNN-based networks, we selected the CNN-PPF network [40], which adopted unique data sample pairs to enhance the hyperspectral pixel data. In addition, feature fusion two-branch CNNs and HRWN networks based on the random walk strategy are also compared. In the network based on the transformer architecture, we contrasted the visual

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:25:56 UTC from IEEE Xplore. Restrictions apply.

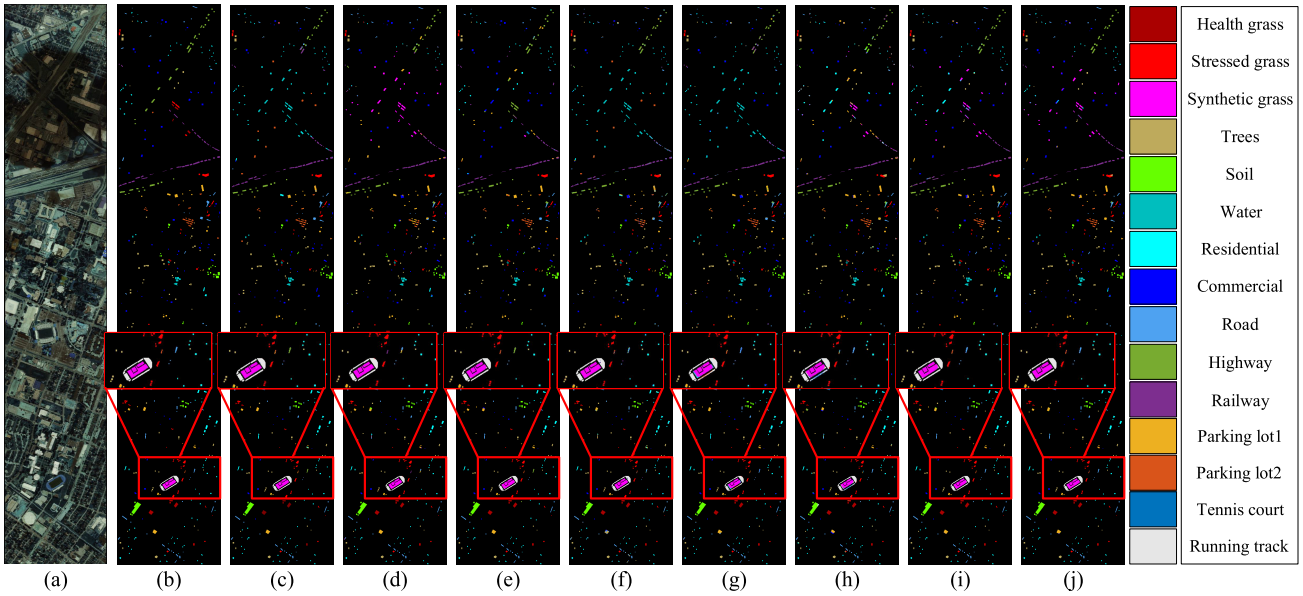


Fig. 3. Classification results of different algorithms on Houston dataset. (a) Pseudo-color map. (b) Ground truth. (c) CNN-PPF. (d) Two-branch. (e) HRWN. (f) ViT. (g) SpectralFormer. (h) DHViT. (i) SSMLP. (j) Ours.

TABLE III
CATEGORIES AND SAMPLE NUMBERS FOR THE MUUFL DATA

Class		Samples Number	
No.	Name	Train	Test
1	Trees	100	23146
2	Mostly grass	100	4170
3	Ground surface	100	6782
4	Dirt	100	1762
5	Road	100	6587
6	Water	100	366
7	Building shadow	100	2133
8	Building	100	6140
9	Sidewalk	100	1285
10	Yellow curb	100	83
11	Cloth panels	100	169
	Total	1100	52632

ViT method [42], which is the foundation of all ViT networks. In addition to this, there are also SpectralFormer networks dedicated to HSI classification and DHViT, a multimodal classification network that fully combines convolutions with ViT. For the network based on the MLP-Mixer architecture, we contrasted an SSMLP-like network for the incremental classification task of hyperspectral [26], and it is worth noting that we removed the experimental setup of the incremental task and adopted the same complete dataset as the other networks for testing. For each algorithm, we all use the same dataset partition, ensure the network to achieve the complete convergence at the same time, and, in order to avoid randomness, adopt different random seed initialized to the network, carry on the experiment many times, and use the average as the final result. In addition, the number of MLP-Mixer blocks used in the experiments was one superparameter, set to a value of 4, as well as the NVIDIA 2080ti graphics card used for training in all experiments.

Specifically, the results of all algorithms are presented in Tables IV–VI, where the best classification performance in each category is bold and the second best is underlined for comparison purposes. Also, the classification visualization results are shown in Figs. 3–5. In addition, Figs. 6–9 show the t-Distributed Stochastic Neighbor Embedding (t-SNE) dimensionality reduction visualization results of HyperMLP on the three datasets. At the same time, we adopted the classification of the general average accuracy (AA), the overall accuracy (OA), and Kappa coefficient (Kappa) as indicators of performance evaluation. Each index is calculated as follows:

$$\mathrm{AA} = \frac{1}{N}\sum_{i=1}^{N}\frac{\mathbf{TP}_i}{\mathbf{S}_i} \tag{23}$$

$$\mathrm{OA} = \mathrm{TP}/S_{\mathrm{all}} \tag{24}$$

where N is the number of data sample categories in this dataset, $\mathbf{TP}_i$ is the sample points correctly classified by each class of samples, and $\mathbf{S}_i$ is the total sample points of each class of samples. TP is the total number of correctly classified samples, while S_{all} is the total number of samples in the dataset.

The number of real samples of each class is defined as $\mathbf{a}_i$, the number of predicted samples of each class is $\mathbf{b}_i$, and the total number of samples is n. Then, p_e is defined as

$$p_e = \frac{\sum_{i=1}^{N}\mathbf{a}_i \cdot \mathbf{b}_i}{n^2}. \tag{25}$$

Then, the formula of Kappa coefficient is defined as

$$\mathrm{Ka} = \frac{\mathrm{OA} - p_e}{1 - p_e}. \tag{26}$$

Also, Kappa coefficient is an index that can measure the performance of classification in imbalanced datasets. It can be

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:25:56 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
ACCURACY COMPARISON WITH OTHER ALGORITHMS ON THE HOUSTON DATASET

No.	CNN-PPF	Two-Branch	HRWN	ViT	SpectralFormer	DHViT	SSMLP	Ours
1	83.00%	83.10%	85.31%	82.72%	81.86%	81.58%	**98.30%**	83.01%
2	84.12%	84.87%	83.79%	80.45%	**100.00%**	78.48%	98.68%	84.49%
3	**100.00%**	**100.00%**	99.05%	99.60%	95.25%	74.26%	99.53%	**100.00%**
4	88.54%	92.14%	92.30%	92.42%	96.12%	90.25%	94.30%	**98.48%**
5	**100.00%**	97.73%	**100.00%**	97.73%	99.53%	99.72%	98.82%	**100.00%**
6	97.20%	68.53%	**97.28%**	95.80%	94.41%	87.41%	89.45%	96.50%
7	83.40%	87.33%	**89.33%**	74.44%	83.12%	84.70%	79.89%	83.44%
8	46.25%	70.75%	93.74%	42.55%	76.73%	**95.35%**	82.41%	80.63%
9	84.04%	84.51%	**88.66%**	65.25%	79.32%	77.15%	79.36%	86.12%
10	56.37%	62.64%	**86.17%**	50.77%	78.86%	42.47%	84.96%	79.54%
11	80.08%	76.47%	92.75%	71.44%	88.71%	79.60%	72.32%	**94.31%**
12	87.42%	91.26%	96.47%	56.00%	87.32%	96.45%	80.55%	**96.73%**
13	82.81%	8.12%	**91.93%**	64.21%	72.63%	79.30%	89.73%	88.07%
14	**100.00%**	**100.00%**	**100.00%**	**100.00%**	**100.00%**	99.60%	99.74%	99.19%
15	98.94%	98.93%	**100.00%**	98.52%	**100.00%**	87.53%	99.34%	99.58%
AA(%)	84.81%	82.70%	90.47%	78.13%	88.91%	83.59%	89.82%	**90.98%**
OA(%)	81.69%	80.42%	89.67%	74.36%	88.01%	82.79%	88.54%	**89.86%**
Kappa	0.803	0.8124	0.8828	0.7243	0.8699	0.8137	0.8761	**0.8903**

TABLE V
ACCURACY COMPARISON WITH OTHER ALGORITHMS ON THE MUUFL DATASET

No.	CNN-PPF	Two-Branch	HRWN	ViT	SpectralFormer	DHViT	SSMLP	Ours
1	88.34%	86.88%	**95.20%**	87.62%	88.83%	92.49%	90.09%	93.15%
2	81.49%	77.19%	84.72%	81.29%	66.62%	80.97%	74.73%	**85.75%**
3	77.25%	**83.57%**	72.93%	59.99%	71.73%	74.26%	76.28%	78.70%
4	93.57%	95.71%	**98.20%**	82.73%	88.47%	90.81%	93.91%	92.12%
5	88.90%	**94.55%**	85.35%	80.12%	84.21%	84.23%	87.23%	92.12%
6	99.18%	61.20%	**100.00%**	84.97%	92.62%	99.68%	98.73%	**100.00%**
7	90.06%	83.54%	94.33%	79.79%	86.45%	94.29%	91.55%	**94.38%**
8	81.12%	94.79%	92.82%	82.88%	83.37%	**95.85%**	95.22%	95.79%
9	72.14%	63.97%	64.36%	71.67%	74.24%	74.41%	80.57%	**86.72%**
10	80.72%	54.22%	85.54%	**97.59%**	89.17%	87.87%	93.94%	90.91%
11	97.63%	94.08%	97.63%	95.86%	96.45%	98.32%	97.48%	**99.16%**
AA(%)	86.40%	80.87%	88.28%	82.24%	83.86%	88.47%	87.36%	**91.84%**
OA(%)	85.53%	86.95%	89.32%	81.23%	83.24%	88.22%	89.06%	**92.73%**
Kappa	0.8122	0.8301	0.8659	0.7564	0.7818	0.8454	0.8344	**0.8926**

TABLE VI
ACCURACY COMPARISON WITH OTHER ALGORITHMS ON THE TRENTO DATASET

No.	CNN-PPF	Two-Branch	HRWN	ViT	SpectralFormer	DHViT	SSMLP	Ours
1	97.13%	89.29%	91.45%	87.35%	96.08%	94.65%	99.79%	**100.00%**
2	92.12%	91.22%	97.83%	81.21%	95.86%	98.42%	**98.96%**	98.66%
3	**98.93%**	83.72%	92.48%	96.79%	95.99%	81.55%	94.12%	93.94%
4	99.10%	98.08%	98.31%	97.42%	97.99%	99.96%	99.33%	**100.00%**
5	96.74%	**100.00%**	99.86%	74.66%	95.25%	99.75%	99.35%	99.99%
6	68.32%	87.27%	83.08%	69.95%	57.76%	86.14%	91.48%	**94.77%**
AA(%)	94.14%	95.55%	96.19%	84.57%	92.37%	96.62%	98.48%	**99.22%**
OA(%)	92.05%	91.61%	93.84%	83.70%	89.82%	93.61%	97.74%	**97.89%**
Kappa	0.9216	0.9403	0.9419	0.7844	0.8982	0.9547	0.9797	**0.9896**

seen from the table that the proposed algorithm achieves the best classification performance on the three indicators on the three datasets. Specifically, on the Houston dataset, compared with the optimal HRWN and SpectralFormer networks in CNN and ViT architectures, each index has a 1% improvement. When compared with the SSMLP network, the AA value is also increased by 1.16%. In the MUUFL dataset, it has a significant result enhancement, which improves the OA value by 3.41%, the AA value by 3.56%, and the kappa value by 2.67% compared with the second-best HRWN network. In TRENTO, compared with the next best SSMLP network, it also has an increase of 0.74% in AA, 0.15% in OA, and 0.99% in kappa.

Correspondingly, in the visualization results of each dataset and algorithm, it can be seen that the result map of the proposed HyperMLP network has the highest similarity with the ground truth. Especially for the MUUFL dataset with complex terrain distribution, the proposed algorithm

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:25:56 UTC from IEEE Xplore. Restrictions apply.

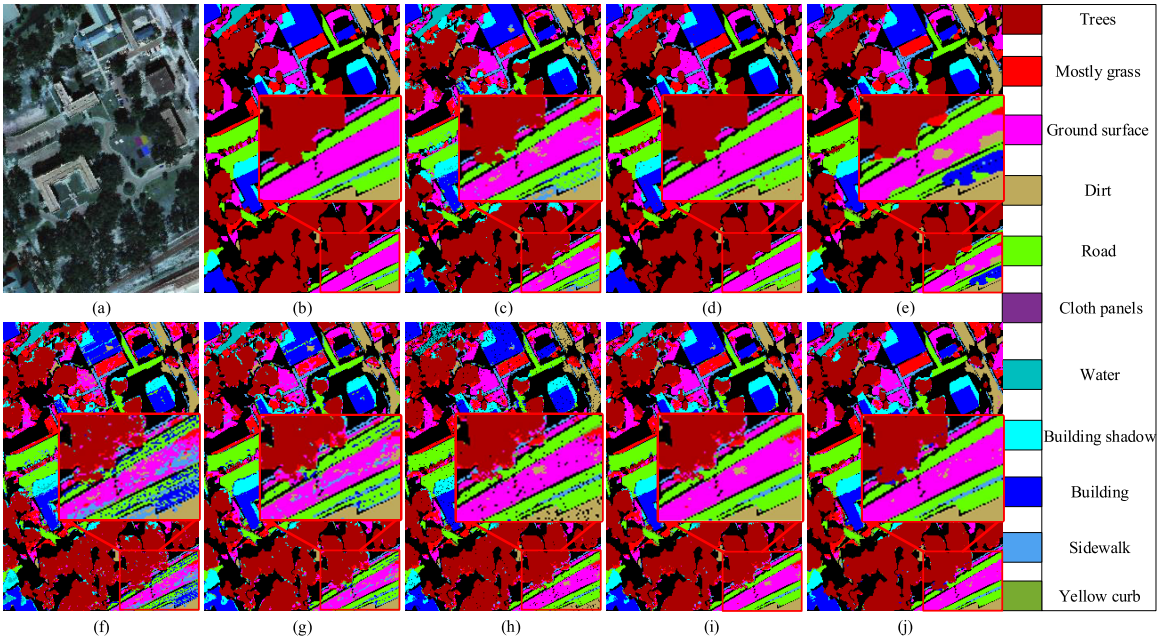


Fig. 4. Classification results of different algorithms on MUUFL Gulfport dataset. (a) Pseudo-color map. (b) Ground truth. (c) CNN-PPF. (d) Two-branch. (e) HRWN. (f) ViT. (g) SpectralFormer. (h) DHViT. (i) SSMLP. (j) Ours.

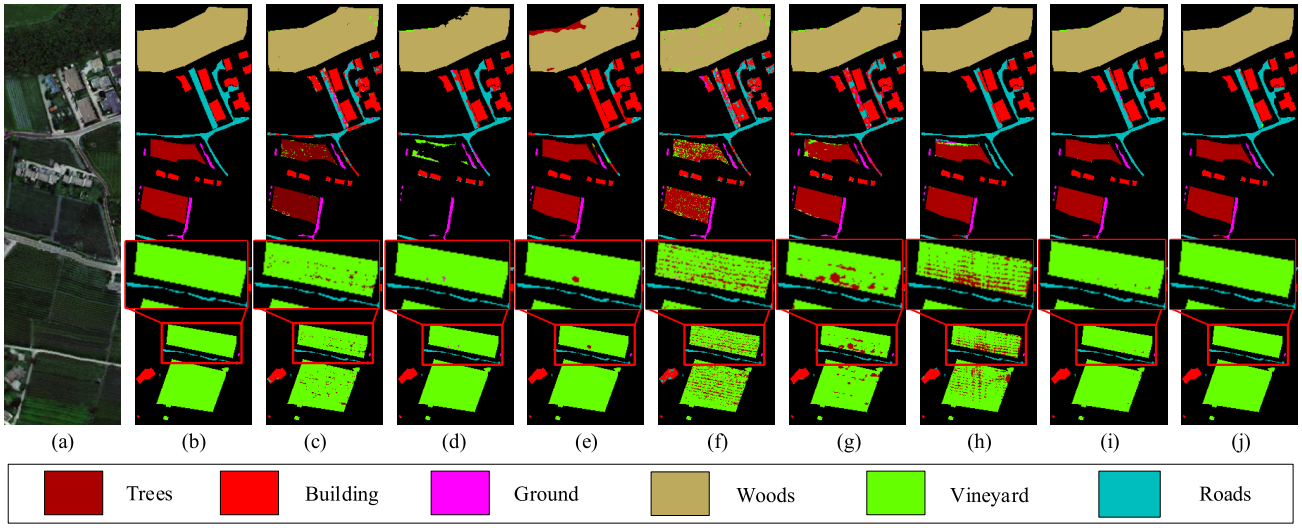


Fig. 5. Classification results of different algorithms on TRENTO dataset. (a) Pseudo-color map. (b) Ground truth. (c) CNN-PPF. (d) Two-branch. (e) HRWN. (f) ViT. (g) SpectralFormer. (h) DHViT. (i) SSMLP. (j) Ours.

also has a smaller noise distribution and more accurate classification at the boundary. Also, it verified the superpixels embedded unsupervised feature extraction for cyberspace ability enhancement, to verify the effectiveness of the proposed module. On the contrary, the best classification performance is also achieved in the Houston dataset, which is relatively distributed and verifies the generalization of the network. On the TRENTO dataset with a large area distribution of land surface, HyperMLP has achieved nearly 100% classification performance, which proves that when the network feature extraction is used to model the spectral information, it also models the spatial information with different distribution well, which verifies the effectiveness of the network.

C. Ablation Experiments

To fully verify the advancement of the innovation proposed in this article, we add each module on the basis of the original MLP network to verify its effectiveness. The results of ablation experiments are shown in Table VII. Specifically, "Basic" represents the original MLP architecture network, "FSM" represents the proposed FSM, "BMS" is the bilateral modulation strategy, "SuperToken" is the superpixel preprocessing feature embedding, and "HyperMLP" is the final proposed network model.

As can be seen from Table VII, the proposed network model achieves the optimal results on the three datasets. Also, the proposed innovation points also have a great improvement

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:25:56 UTC from IEEE Xplore. Restrictions apply.

on the original basic network. The proposed FSM module can effectively enhance the generalization of the network and improve the information interaction between features. This advantage is most obvious on the TRENTO and Houston datasets, which both have about 3%–4% improvement on the three evaluation indicators. For the BMS module, the feature fusion between different data can effectively correct the deviation of feature extraction in the later stage. For the MUUFL dataset with more data categories and complex distribution, this module makes the network increase the OA, AA, and Kappa coefficient by 2.18%, 2.18%, and 5.16%, respectively. The additional data preprocessing scheme superpixel SuperToken has a significant promotion for the three datasets. The prior image spatial aggregation information it provides can more effectively help the network to converge, combine with the original data, and improve the spatial perception ability of the network.

D. Effect of Different Input Patches

In this section, we verify the effect of different input sizes on the final output of the network. In the network, there are two parts of input, which are HSI input and LiDAR image input. For LiDAR images, the input information is relatively single, so we fixed its input size as 15×15, and on this basis, we explored the size of HSIs. Specifically, we tested 7×7, 9×9, 11×11, 13×13, and 15×15 size inputs to verify the best size input on different datasets.

The specific results are recorded in Table VIII. In detail, on the TRENTO dataset, the best classification results are achieved when the HSI input size is 7×7, and as we gradually increase the data input, the classification results gradually decrease. In the Houston and TRENTO datasets, the best classification results are obtained when the input size is 11×11, and the final results are affected by the input size being too small or too large. In the TRENTO dataset, the number of terrain categories is small and the overall distribution is relatively uniform, showing a block distribution. In this case, the network has a strong ability to extract spatial features. When the input size is too large, additional boundary pixels will be introduced, which will affect the classifier. On the contrary, in the complicated terrain category, distribution of discrete Houston and MUUFL datasets, the input size is too small, which will lead to the network not fully modeling of complex space information, but the size will introduce more noise, so the selection of medium size as input can get the best classification performance.

E. Effect of the Number of Initial Pixels in Superpixel Embedding

In HyperMLP, we use the unique unsupervised superpixel segmentation method as additional data embedding, but in the ERS, there is a key parameter σ, which affects the result of the superpixel, so in this section, we explore the impact of this hyperparameter on the final network classification. Specifically, the three most clearly differentiated initial point settings were used in the a priori superpixel task [43], [44], [45], and thus, we set the σ values of the three data to 100, 50, and 20 to test the experimental results at different initial pixels.

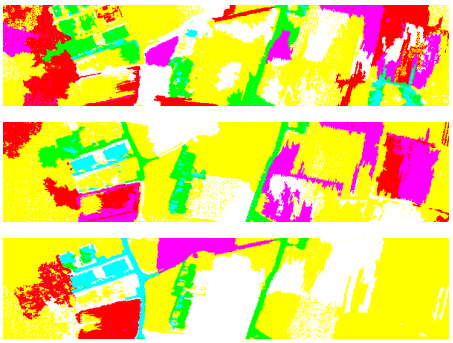


Fig. 6. Visualization of different initial pixel points for superpixel in the TRENTO dataset.

Fig. 7. Visualization of different initial pixel points for superpixel in the MUUFL dataset.

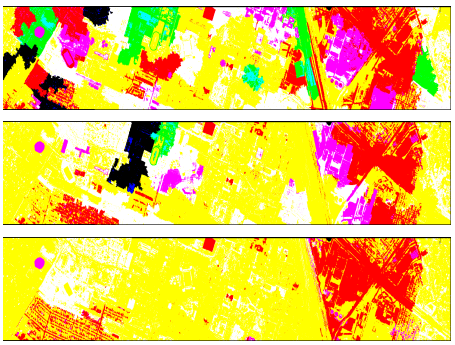


Fig. 8. Visualization of different initial pixel points for superpixel in the Houston dataset.

Specifically, we set σ to three values of 100, 50, and 20 on the three data in order to test the experimental results under different initial pixels. In addition, for easy observation, the superpixel segmentation results under different initial pixels are plotted, and it is worth noting that we selected the same three channels for visualization. The specific visual results as shown in Figs. 6–8 and classification results as shown in Figs. 12–14.

The influence of different initial points on the final result can be seen from the superpixel segmentation result. When the initial number of pixels is 20, the overall segmentation result is relatively whole, and the departments with obvious boundaries are initially segmented, but the details still cannot

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:25:56 UTC from IEEE Xplore. Restrictions apply.

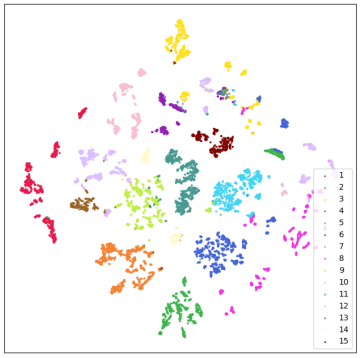


Fig. 9. Low-dimensional visualization results of t-SNE method for high-dimensional features of Houston dataset.

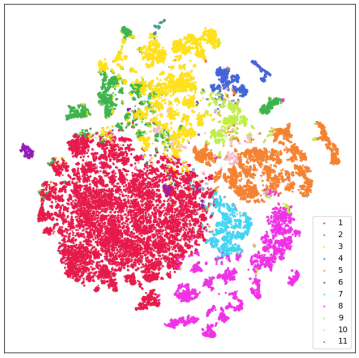


Fig. 10. Low-dimensional visualization results of t-SNE method for high-dimensional features of TRENTO dataset.

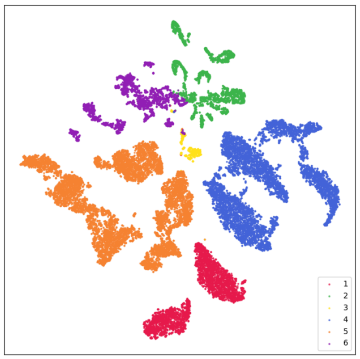


Fig. 11. Low-dimensional visualization results of t-SNE method for high-dimensional features of MUUFL Gulfport dataset.

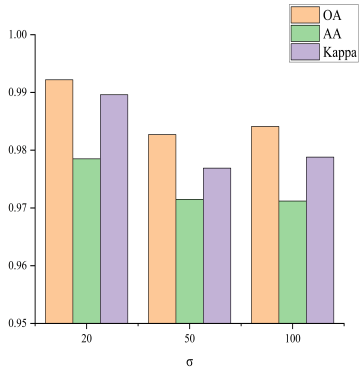


Fig. 12. Results of different pixel initialization parameters σ under the TRENTO dataset.

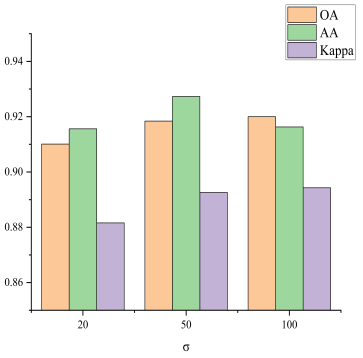


Fig. 13. Results of different pixel initialization parameters σ under the MUUFL dataset.

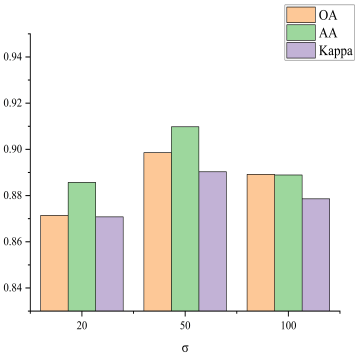


Fig. 14. Results of different pixel initialization parameters σ under the Houston dataset.

be effectively distinguished. When an initial pixel is too large, detail boundary can be extracted and, at the same time, compared with small pixel points, have a variety of categories. From the final results, for the TRENTO dataset, a small initial number of pixels gives the best classification performance, and for the MUUFL dataset and the Houston dataset, a moderate number of 50 pixels is better. The results coincided with the results corresponding to the input sizes. For the TRENTO dataset, the categories are few and the distribution is simple, and the more holistic superpixel segmentation helps the final result more. For the other two more complex datasets, too few initial pixels will lead to less spatial information embedding, and too many will lead to wrong segmentation, which will affect the final classification performance.

TABLE VII

RESULTS OF ABLATION EXPERIMENTS FOR THE THREE PROPOSED MODULES

Dateset	Index	Basic	Basic +FSM	Basic +BMS	Basic +SuperToken	HyperMLP
TRENTO	OA	95.00%	98.06%	97.79%	98.36%	**99.04%**
	AA	91.85%	97.67%	96.92%	97.44%	**97.89%**
	kappa	0.9332	0.9784	0.9704	0.9781	**0.9872**
MUUFL	OA	89.00%	91.12%	91.18%	90.36%	**91.32%**
	AA	88.44%	90.65%	90.62%	91.87%	**93.01%**
	kappa	0.8318	0.8863	0.8834	0.8736	**0.8875**
Houston	OA	85.68%	88.23%	86.61%	88.16%	**89.04%**
	AA	84.51%	89.92%	88.63%	89.32%	**90.98%**
	kappa	0.874	0.8727	0.8553	0.8718	**0.8798**

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:25:56 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
EFFECT OF INPUT PATCH SIZE ON THE FINAL CLASSIFICATION RESULTS IN DIFFERENT DATASETS

Patch Size	Dataset	OA	AA	Kappa
7×7	TRENTO	**99.04%**	**97.89%**	**0.9872**
9×9	TRENTO	98.98%	97.52%	0.9864
11×11	TRENTO	98.87%	97.51%	0.9849
13×13	TRENTO	98.32%	96.87%	0.9805
15×15	TRENTO	97.95%	96.51%	0.9737

Patch Size	Dataset	OA	AA	Kappa
7×7	MUUFL	90.84%	92.30%	0.8796
9×9	MUUFL	91.21%	92.45%	0.8838
11×11	MUUFL	**91.32%**	**93.01%**	**0.8857**
13×13	MUUFL	90.40%	91.80%	0.8758
15×15	MUUFL	90.04%	90.20%	0.8687

Patch Size	Dataset	OA	AA	Kappa
7×7	Houston	88.55%	90.83%	0.8746
9×9	Houston	88.85%	90.79%	0.8778
11×11	Houston	**89.04%**	**90.98%**	**0.8798**
13×13	Houston	88.69%	90.52%	0.8759
15×15	Houston	88.61%	90.50%	0.8752

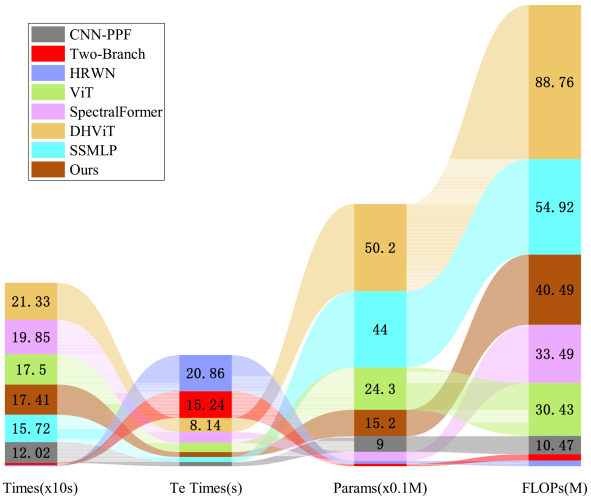


Fig. 15. Complexity of different algorithms on the TRENTO dataset, where Tr Times represents the training time and Te Times represents the testing time.

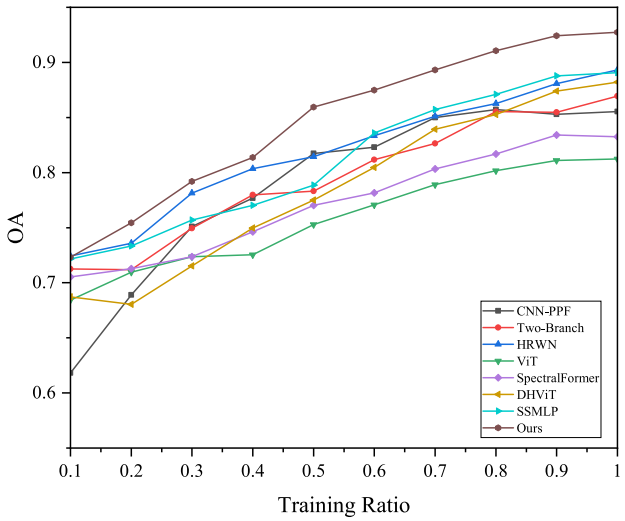


Fig. 16. Influence of different proportions of training samples on the final classification.

F. Complexity Comparison of Different Algorithms

This section investigates the complexity between different comparison algorithms. Specifically, the training time, testing time, number of parameters, and floating-point operations per second (FLOPs) of different algorithms are tested on the TRENTO dataset. The specific results are shown in Fig. 15. In order to put different data in one figure for easy observation, we adopted different proportions for different parameters, so the broken lines in the figure only represent the change trend and comparison, and the specific adjustment proportions are illustrated in the legend in the figure. In addition, for the sake of comparison, several algorithms with higher complexity are represented by larger proportions, and the corresponding values are also marked in the figure. As can be seen from the figure, compared to the traditional CNN-based network, the ViT and MLP-based networks have quite high training and testing time, indicating that they need more training rounds to achieve convergence. In terms of the number of parameters, the network of ViT architecture also has more parameters than the network of CNN architecture, mainly due to the complex attention interaction calculation in ViT architecture, which is also in line with the theoretical expectation in the specific experimental results. In addition, compared with DHViT and SSMLP networks, the proposed HyperMLP network has fewer network parameters and achieves better classification performance. Although HyperMLP has a larger time complexity and parameter number than some CNN-based networks, the significant improvement in accuracy obtained by HyperMLP is considered acceptable.

G. Different Proportions of Training Samples

In this section, we randomly selected the number of different proportions from the MUUFL data training set in ten training runs (10%–100%), Then, the OA of different algorithms is compared, and the specific results are shown in Fig. 16. It is worth noting that for the three validated indicators (OA, AA, and Kappa), there are similar change trends, so we only show the change trend of OA in the figure. It can be seen from the figure that for different proportions of training set data, the proposed algorithm has excellent classification performance. When the amount of data is small, the proposed algorithm has no great performance degradation, and when the amount of data increases, HyperMLP reflects the powerful feature extraction and classification ability. In addition, the stability of the proposed algorithm provides the possibility for its application in the case of few samples.

V. CONCLUSION

This article proposes HyperMLP, a novel multimodality classification network based on the MLP-Mixer framework for hyperspectral and LiDAR elevation imagery. Specifically, this network utilizes superpixel data embedding to enhance its ability to capture spatial features. Furthermore, a special FSM is employed to improve the spatial feature extraction

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:25:56 UTC from IEEE Xplore. Restrictions apply.

level of the network. In addition, BMS improves the network characteristics of global awareness, strengthening the contact and communication between multimodal data. At the same time, on three benchmark datasets, the proposed method achieves excellent classification performance compared to advanced networks with multiple architectures. A separate section of this article explores the initial superpixel pixel points, as well as the impact of different input sizes on the results of the network. Generally, this article presents a method of fusion classification for hyperspectral and LiDAR images based on MLP architecture for the first time, providing new ideas and directions for HSI classification.

REFERENCES

[1] M. Fauvel, Y. Tarabalka, J. A. Benediktsson, J. Chanussot, and J. C. Tilton, "Advances in spectral–spatial classification of hyperspectral images," *Proc. IEEE*, vol. 101, no. 3, pp. 652–675, Mar. 2013.

[2] P. Ghamisi et al., "Advances in hyperspectral image and signal processing: A comprehensive overview of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 37–78, Dec. 2017.

[3] P. Ghamisi, J. Plaza, Y. Chen, J. Li, and A. J. Plaza, "Advanced spectral classifiers for hyperspectral images: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 1, pp. 8–32, Mar. 2017.

[4] L. He, J. Li, C. Liu, and S. Li, "Recent advances on spectral–spatial hyperspectral image classification: An overview and new guidelines," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1579–1597, Mar. 2018.

[5] J. Ngiam, A. Khosla, M. Kim, J. Nam, H. Lee, and A. Y. Ng, "Multimodal deep learning," in *Proc. 28th Int. Conf. Mach. Learn. (ICML)*, 2011, pp. 689–696.

[6] W. Li, Y. Gao, M. Zhang, R. Tao, and Q. Du, "Asymmetric feature fusion network for hyperspectral and SAR image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 10, pp. 8057–8070, Oct. 2023, doi: 10.1109/TNNLS.2022.3149394.

[7] Z. Chen, H. Pu, B. Wang, and G.-M. Jiang, "Fusion of hyperspectral and multispectral images: A novel framework based on generalization of pan-sharpening methods," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 8, pp. 1418–1422, Aug. 2014.

[8] B. Arad and O. Ben-Shahar, "Sparse recovery of hyperspectral signal from natural RGB images," in *Proc. 14th Eur. Conf. Comput. Vis.* Amsterdam, The Netherlands. Cham, Switzerland: Springer, Oct. 2016, pp. 19–34.

[9] M. Zhang, W. Li, Y. Zhang, R. Tao, and Q. Du, "Hyperspectral and LiDAR data classification based on structural optimization transmission," *IEEE Trans. Cybern.*, vol. 53, no. 5, pp. 3153–3164, May 2023.

[10] J. Li, Y. Liu, R. Song, Y. Li, K. Han, and Q. Du, "Sal^{2}RN: A spatial–spectral salient reinforcement network for hyperspectral and LiDAR data fusion classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–14, 2023, Art. no. 5500114.

[11] H. Lee and H. Kwon, "Going deeper with contextual CNN for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 26, no. 10, pp. 4843–4855, Oct. 2017.

[12] X. Xu, W. Li, Q. Ran, Q. Du, L. Gao, and B. Zhang, "Multisource remote sensing data classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 937–949, Feb. 2018.

[13] R. Hang, Z. Li, P. Ghamisi, D. Hong, G. Xia, and Q. Liu, "Classification of hyperspectral and LiDAR data using coupled CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4939–4950, Jul. 2020.

[14] S. Mohla, S. Pande, B. Banerjee, and S. Chaudhuri, "FusAtNet: Dual attention based spectro-spatial multimodal fusion network for hyperspectral and LiDAR classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 92–93.

[15] X. He, Y. Chen, and Z. Lin, "Spatial–spectral transformer for hyperspectral image classification," *Remote Sens.*, vol. 13, no. 3, p. 498, Jan. 2021.

[16] K. Han et al., "A survey on vision transformer," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 1, pp. 87–110, Jan. 2023.

[17] S. Khan, M. Naseer, M. Hayat, S. W. Zamir, F. S. Khan, and M. Shah, "Transformers in vision: A survey," *ACM Comput. Surv.*, vol. 54, no. 10, pp. 1–41, Jan. 2022.

[18] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022, Art no. 5518615.

[19] Z. Xue, X. Tan, X. Yu, B. Liu, A. Yu, and P. Zhang, "Deep hierarchical vision transformer for hyperspectral and LiDAR data classification," *IEEE Trans. Image Process.*, vol. 31, pp. 3095–3110, 2022.

[20] Z. Liu et al., "Swin Transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[21] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 16000–16009.

[22] I. Tolstikhin et al., "MLP-mixer: An all-MLP architecture for vision," in *Proc. 35th Conf. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 24261–24272.

[23] S. Chen, E. Xie, C. Ge, R. Chen, D. Liang, and P. Luo, "CycleMLP: A MLP-like architecture for dense prediction," 2021, *arXiv:2107.10224*.

[24] T. Yu, X. Li, Y. Cai, M. Sun, and P. Li, "S2-MLP: Spatial-shift MLP architecture for vision," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2022, pp. 297–306.

[25] H. Touvron et al., "ResMLP: Feedforward networks for image classification with data-efficient training," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 4, pp. 5314–5321, Apr. 2023.

[26] Y. Sun, B. Liu, R. Wang, P. Zhang, and M. Dai, "Spectral–spatial MLP-like network with reciprocal points learning for open-set hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–18, 2023, Art. no. 5513218.

[27] J. Guo et al., "Hire-MLP: Vision MLP via hierarchical rearrangement," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 816–826.

[28] Y. Shao, J. Liu, J. Yang, and Z. Wu, "Spatial–spectral involution MLP network for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 9293–9310, 2022.

[29] X.-J. Tang, X. Liu, P.-F. Yan, B.-X. Li, H.-Y. Qi, and F. Huang, "An MLP network based on residual learning for Rice hyperspectral data classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[30] M. Lin, W. Jing, D. Di, G. Chen, and H. Song, "Multi-scale U-shape MLP for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[31] V. Jampani, D. Sun, M.-Y. Liu, M.-H. Yang, and J. Kautz, "Superpixel sampling networks," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 352–368.

[32] R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, "SLIC superpixels compared to state-of-the-art superpixel methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 11, pp. 2274–2282, Nov. 2012.

[33] M.-Y. Liu, O. Tuzel, S. Ramalingam, and R. Chellappa, "Entropy rate superpixel segmentation," in *Proc. CVPR*, Jun. 2011, pp. 2097–2104.

[34] S. Wang, H. Lu, F. Yang, and M.-H. Yang, "Superpixel tracking," in *Proc. Int. Conf. Comput. Vis.*, Nov. 2011, pp. 1323–1330.

[35] A. Levinshtein, A. Stere, K. N. Kutulakos, D. J. Fleet, S. J. Dickinson, and K. Siddiqi, "TurboPixels: Fast superpixels using geometric flows," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 31, no. 12, pp. 2290–2297, Dec. 2009.

[36] M. Van den Bergh, X. Boix, G. Roig, B. De Capitani, and L. Van Gool, "SEEDS: Superpixels extracted via energy-driven sampling," in *Proc. 12th Eur. Conf. Comput. Vis. (ECCV)*, Florence, Italy. Berlin, Germany: Springer, Oct. 2012, pp. 13–26.

[37] J. Chen, Z. Li, and B. Huang, "Linear spectral clustering superpixel," *IEEE Trans. Image Process.*, vol. 26, no. 7, pp. 3317–3330, Jul. 2017.

[38] J. Wang, W. Li, Y. Gao, M. Zhang, R. Tao, and Q. Du, "Hyperspectral and SAR image classification via multiscale interactive fusion network," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 12, pp. 10823–10837, Dec. 2023, doi: 10.1109/TNNLS.2022.3171572.

[39] Y. Ning, J. Peng, Q. Liu, Y. Huang, W. Sun, and Q. Du, "Contrastive learning based on category matching for domain adaptation in hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–14, 2023, Art. no. 5301814.

[40] W. Li, G. Wu, F. Zhang, and Q. Du, "Hyperspectral image classification using deep pixel-pair features," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 844–853, Feb. 2017.

[41] X. Zhao et al., "Joint classification of hyperspectral and LiDAR data using hierarchical random walk and deep CNN architecture," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7355–7370, Oct. 2020.

[42] A. Dosovitskiy et al., "An image is worth 16 × 16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:25:56 UTC from IEEE Xplore. Restrictions apply.

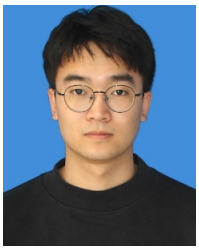
[43] J. Li, Y. Ma, R. Song, B. Xi, D. Hong, and Q. Du, "A triplet semisupervised deep network for fusion classification of hyperspectral and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022, Art. no. 5540513.

[44] Y. Huang et al., "Two-branch attention adversarial domain adaptation network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022, Art. no. 5540813.

[45] Y. Huang et al., "Cross-scene wetland mapping on hyperspectral remote sensing images using adversarial domain adaptation network," *ISPRS J. Photogramm. Remote Sens.*, vol. 203, pp. 37–54, Sep. 2023.

Jiaojiao Li (Senior Member, IEEE) received the B.E. degree in computer science and technology, the M.S. degree in software engineering, and the Ph.D. degree in communication and information systems from Xidian University, Xi'an, China, in 2009, 2012, and 2016, respectively.

She was an Exchange Ph.D. Student with Mississippi State University, Starkville, MS, USA, supervised by Dr. Qian Du. She is currently an Associate Professor and a Doctoral Supervisor with the School of Telecommunication, Xidian University. Her research interests include hyperspectral remote sensing image analysis and processing, and pattern recognition.



Yuzhe Liu received the B.E. degree in telecommunications engineering from Xidian University, Xi'an, China, in 2021, where he is currently pursuing the M.S. degree in information and communication engineering with the State Key Laboratory of Integrated Service Network.

His research interests include hyperspectral image process, machine learning, and deep learning.

Rui Song (Member, IEEE) received the Ph.D. degree in signal and information processing from Xidian University, Xi'an, China, in 2009.

He is currently a Professor and a Ph.D. Advisor with the State Key Laboratory of Integrate Service Network, School of Telecommunications, Xidian University. His research interests include image and video coding algorithms and VLSI architecture design, intelligent image processing, and understanding and reconstruction of 3-D scene.

Wei Liu received the master's degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2011.

She is currently working with the State Key Laboratory of Geo-Information Engineering, Wuhan University. Her research interests include real-time processing of unmanned aerial vehicle (UAV) images and satellite images.

Yunsong Li (Member, IEEE) received the M.S. degree in telecommunication and information systems and the Ph.D. degree in signal and information processing from Xidian University, Xi'an, China, in 1999 and 2002, respectively.

In 1999, he joined the School of Telecommunications Engineering, Xidian University, where he is currently a Professor. He is also the Director of the Image Coding and Processing Center, State Key Laboratory of Integrated Service Networks, Xidian University. His research interests focus on image and video processing, hyperspectral image processing, and high-performance computing.

Qian Du (Fellow, IEEE) received the Ph.D. degree in electrical engineering from the University of Maryland at Baltimore County, Baltimore, MD, USA, in 2000.

She is currently a Bobby Shackouls Professor with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS, USA. Her research interests include hyperspectral remote-sensing image analysis and applications, pattern classification, data compression, and neural networks.

Dr. Du is a fellow of SPIE—International Society for Optics and Photonics. She was a recipient of the 2010 Best Reviewer Award from the IEEE Geoscience and Remote Sensing Society (GRSS). She was the Co-Chair of the Data Fusion Technical Committee of the IEEE GRSS from 2009 to 2013. She was the Chair of the Remote Sensing and Mapping Technical Committee of the International Association for Pattern Recognition from 2010 to 2014. She was the General Chair for the fourth IEEE GRSS Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing held in Shanghai, China, in 2012. She served as an Associate Editor for the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (JSTARS), the JOURNAL OF APPLIED REMOTE SENSING, and the IEEE SIGNAL PROCESSING LETTERS. Since 2016, she has been the Editor-in-Chief of the IEEE JSTARS.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 15:25:56 UTC from IEEE Xplore. Restrictions apply.